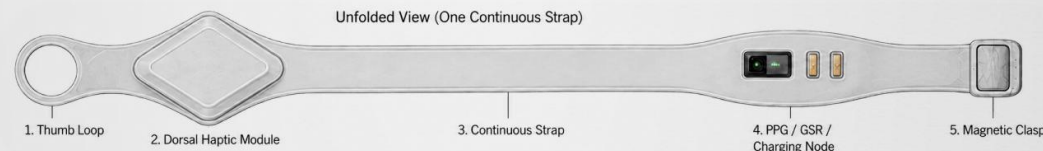




Wearable Wristband Based on Vision-Language Model Environmental Interpretation

Assisting Independent Mobility
for Low-Vision Users

Theme Practice



USER RESEARCH

Heideng

A professional stand-up comedian, he has a lot of fans in China.

Basic Profile

Occupation	Stand-up comedian and rare-disease advocate
Vision	Stargardt disease, with visual acuity of approximately 0.02. Severe central vision loss; mainly perceives blurred outlines
Digital literacy	Photographs distant text and enlarges it on his phone. Uses OCR and text-to-speech to access visual information.
Mobility	Memorises metro routes, stations and transfers in advance. Moves closer to signs or uses his phone to recognise text.
Personality	Independent, candid, humorous and observant; dislikes unnecessary assistance

GOALS

This user's goals were developed from publicly available interviews and media reports.

1. Maintain independence and control over everyday mobility.
2. Access the most important environmental information quickly.
3. Reduce repeated phone-based interactions.
4. Move efficiently when travelling between performances.
5. Receive assistance without being controlled, labelled or medicalised by the product.
6. Preserve personal style and a dignified social identity.

DEMANDS DIG

Functional Needs

- Clear directional guidance.
- Timely alerts for obstacles, steps, turns and route deviation.
- Recognition of seat numbers, door signs, platforms and essential signage.
- Interaction that does not require continuous screen attention.
- Haptic feedback that remains perceptible in noisy environments.

Emotional & Social Needs

- Avoid an overly medical appearance.
- Avoid unnecessarily revealing disability or private information.
- Fit naturally with personal style, clothing

IDEATION

Traditional navigation apps are not enough, so we need to use first-person-view image recognition of the surroundings.

We need wearable devices and enables human-computer interaction via multi-sensory signals beyond vision.

It should function like an everyday accessory — natural to use, portable, and even somewhat fashionable.



Camera clip + voice module

Sunglasses is existing daily eyewear, no extra hardware added. The portable camera clip captures the user's first-person perspective, while the voice module supports real-time voice interaction.

AI processes visual information triggered by voice-input prompts.



This portable strap-form design occupies minimal palm-tactile capacity.

Why reject minimalist ring or fitness-band forms?

While they minimize medical-device aesthetics, wrists are poor for clear directional vibration feedback. Research shows the back-of-hand has duller tactile perception and prevents excessive haptic stimulation.

Why not a full-enclosure glove structure?

Tactile sensation is extremely valuable for people with visual impairments, we need to leave the fingertip of the index finger (the most sensitive) exposed for braille reading. Meanwhile, the palm should also remain uncovered for object perception and grasping.

Why reject the strap-style cut-out glove structure?

The integrated form offers limited flexibility and cannot fit all users with a single size. In addition, thin straps are prone to damage.

Heideng is a highly independent person with low vision. He works, travels between cities and uses public transport independently, relying on route memory and digital accessibility tools. He does not need technology to take control on his behalf; he needs timely and reliable information that helps him remain in control.

System Design

00 WHY THIS SCENARIO?

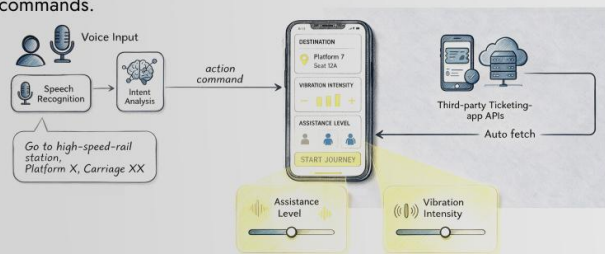
Common outdoor scenarios are supported by Google Maps through external API navigation inputs. **By contrast**, underground parking lots and subway stations depend heavily on way-finding signs and landmarks for directional guidance, creating new obstacles for low-vision users.

This Project target GPS-unavailable environments, solution integrates **first-person vision data + AI + wearable multi-sensory** hardware to enable independent mobility for low-vision people.

01 SET

Before-Trip

- Voice-based destination & requirement setup. Data can also be auto-fetched via third-party ticketing-app APIs.
- Personalisation: adjust assistance level and vibration intensity using voice commands.



This scenario is designed exclusively for low-vision users with residual eyesight (see previous user case).



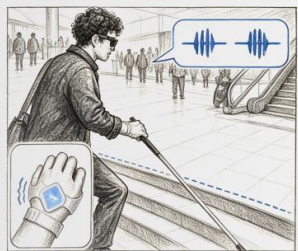
01 SET



02 SENSE



03 TURN



04 ALERT



05 IDENTIFY

When visual recognition detects obstacles (stairs, sudden pedestrians or objects) and triggers an alert to tell the user to stop.

Voice confirmation will be triggered when the user approaches the destination.



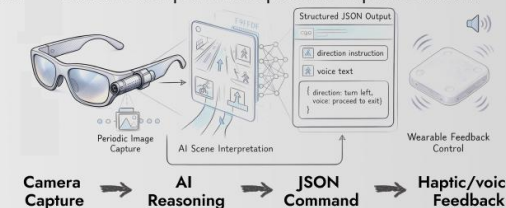
06 ARRIVE

02 SENSE - 03 TURN

During-Trip

Upon entering GPS-denied areas, the camera integrated into the sunglass temple activates and captures images at periodic intervals.

AI analyses the data and outputs structured JSON data with direction instructions and voice text. The wearable device receives signals and drives electronic components to produce haptic feedback.

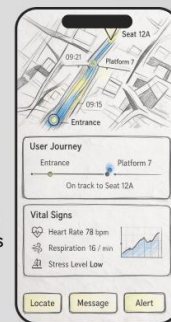


06 ARRIVE

After-Trip

Upon arrival, navigation task ends. Trajectory logs are saved in the backend and shown on mobile for family/caregiver access.

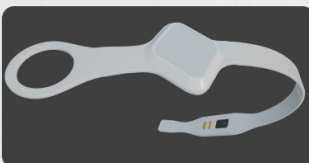
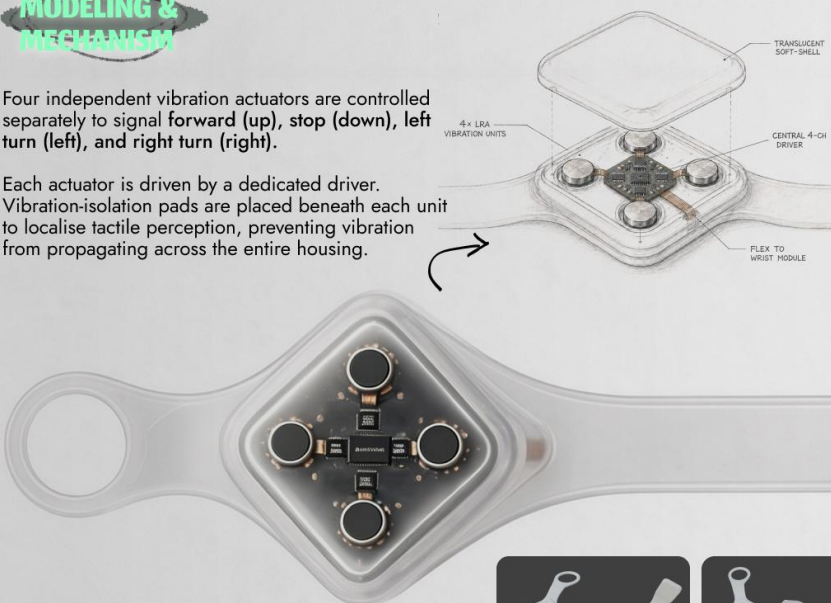
The device also monitors the user's physiological status in real-time. Alerts will be sent to family members if abnormal vital signs are detected.



MODELING & MECHANISM

Four independent vibration actuators are controlled separately to signal forward (up), stop (down), left turn (left), and right turn (right).

Each actuator is driven by a dedicated driver. Vibration-isolation pads are placed beneath each unit to localise tactile perception, preventing vibration from propagating across the entire housing.



EXPERIMENT & CODING

Due to component availability constraints, this experiment was conducted solely on the four-directional haptic vibration module

Current component choices only satisfy basic functionality. Do not fit wearable size

Code testing uses VS Code with the PlatformIO extension, where serial commands simulate large language model inputs.



MCU
ESP32-C3 SuperMini



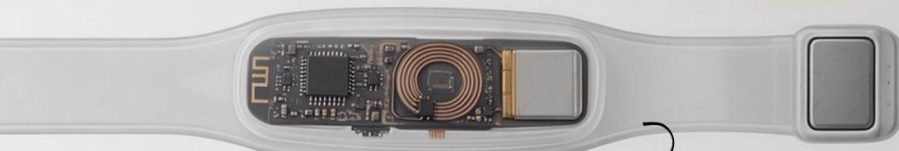
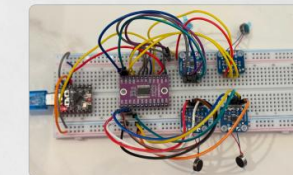
I2C Multiplexer
PCA9548A



Haptic Drivers
DRV2605L X4



Z-axis Vibration Unit
LRA-0825 X4



The wrist-unit consists of two layers.

The upper layer houses the main controller and wireless charging module.

The lower layer (skin-facing side) accommodates two physiological sensors (PPG + GSR).

